CASE STUDY BREAKDOWN

Cooper Industries

The Nicholson File Acquisition

How to Value a Merger and Design the Share Exchange

How to use this guide (read first)

This is a step-by-step playbook for a **merger structuring** case — the type where a buyer (Cooper) wants to acquire a target (Nicholson) by swapping shares, and you must work out **how many buyer shares to give per target share**. It is written for a non-finance student: every method is explained as **Why** → **How** → **Steps** → **Watch-outs**, using the exact numbers from your solution. If you learn the four methods and the one big idea ("don't dilute the buyer's earnings"), you can solve any case of this family.

How to recognise this case type in the exam

You are looking at a **Cooper-type (merger / exchange-ratio)** problem whenever you see these signals:

- One company wants to **acquire another by issuing its own shares** (a "stock swap" or "exchange offer"), not paying cash.
- The question asks for a **swap ratio / exchange ratio** — how many acquirer shares per target share.
- You are told about **different shareholder groups** (family, promoters, speculators) with different motives.
- There is a concern about **diluting earnings per share (EPS)** or **book value** after the deal.
- Contrast this with: *Kohler* = valuing a **private** company for a fair-value lawsuit; *Stanley Black & Decker* = testing whether **synergies** beat the premium. Different questions, different tools.

1 The Situation — What Is Actually Happening

The players

Party	Who they are	What they want
Cooper Industries	The acquirer. A larger industrial company looking to grow by acquisition.	Buy control of Nicholson — but without overpaying or hurting its own share price.
Nicholson File Co.	The target. A hand-tool maker with a strong brand and distribution, but under-performing.	A fair price; the family wants control, security, and no tax bill on the sale.
VLN Corp.	A rival bidder also chasing Nicholson.	Win Nicholson for itself — Cooper must beat this offer.
Nicholson family / promoters	Hold ~117,000 shares. Founders.	Keep some control and freedom; avoid tax; a good long-term partner.
Speculators / arbitrageurs	Bought shares hoping for a bid.	The highest short-run price, in cash.

The core problem in one line

Cooper wants to give Nicholson shareholders **Cooper shares** in exchange for their **Nicholson shares**. The whole case is deciding the **swap ratio**: how many Cooper shares to hand over for each Nicholson share. Offer too few → Nicholson shareholders reject it. Offer too many → Cooper hurts its own shareholders (dilution). The right answer must satisfy **both sides at once**.

What is a "swap ratio" (exchange ratio)? — the single most important idea

Think of it as an exchange rate between two currencies. If the swap ratio is **2.0**, then **1 Nicholson share = 2 Cooper shares**. A Nicholson holder with 100 shares would receive 200 Cooper shares. The entire negotiation is about setting this one number fairly — and the four methods below are simply four different **lenses** for deciding what is fair.

The four lenses (your whole toolkit)

1. **DCF (Discounted Cash Flow)** — what is Nicholson truly *worth* on its own, in cash terms? Sets a **floor**.
2. **Swap based on EPS** — the most shares Cooper can give *without reducing its own earnings per share*. Sets a **ceiling** for Cooper.
3. **Swap based on Book Value** — exchange in proportion to each share's accounting net worth. A **cross-check**.
4. **Shareholder-group analysis** — the human/negotiation lens: what each group needs to say yes.

2 Method 1 — DCF Valuation (What Nicholson Is Really Worth)

WHY use this method

Before you swap shares, you need to know what Nicholson is **intrinsically worth** — not its stock price (which is noisy and pushed up by the takeover rumour), but the cash the business will actually generate. DCF answers: *"if I owned this whole company and kept all its future cash, what is that worth today?"* This gives Cooper a **floor** — a value it can defend and shouldn't go below.

THE BIG IDEA behind DCF (plain English)

> £1 next year is worth **less** than £1 today, because today's pound can be invested. So DCF projects the company's future cash for ~10 years, then **shrinks ("discounts")** each future year back to today's value using a rate that reflects risk. Add up all those shrunk cash flows → that is the company's value today. Divide by the number of shares → value per share.

HOW to solve it — step by step

Step 1 — Project revenue

Start from the last actual year (1971 revenue = $55.3MM) and grow it. Your solution grows revenue at **6% per year** for 10 years (1972–1981). Formula each year: `Revenue = prior year × 1.06`.

Step 2 — Subtract costs to get operating profit (NOP)

From revenue subtract COGS (cost of goods, improving from 69% → 65% of sales), SG&A (22% → 19%), Depreciation ($2.1MM fixed), and other deductions ($0.2MM). What's left is **Net Operating Profit (NOP)**.

Step 3 — Subtract tax to get profit after tax (PAT)

Tax = 40% of (NOP − interest). Subtract it: `PAT = NOP − Tax`. This is the accounting profit.

Step 4 — Convert profit to CASH (Free Cash Flow to Equity)

Profit is not cash. Adjust it: **add back Depreciation** (it was subtracted but no cash left), **subtract the increase in Working Capital** (cash tied up in inventory/receivables as the firm grows), and **subtract Capital Expenditure** (cash spent on equipment). Result = **FCFE**, the real cash available to shareholders.

Step 5 — Pick the discount rate (Cost of Equity)

Because we discount cash to *equity*, we use the **Cost of Equity (Ke)**. Your solution uses the **Dividend Discount Model**: `Ke = D1/P + g` = (1.4×1.05)/27.5 + 0.05 ≈ **10.3%** (rounded to 10% for discounting).

Step 6 — Discount every year back to today

For each year multiply its FCFE by a **PV factor** = `1 / (1+10%)^year`. Year 1 factor = 0.909, Year 2 = 0.826, and so on — further years shrink more.

Step 7 — Add a Terminal Value

The company keeps generating cash after year 10. Capture that as a **Terminal Value**: `final-year FCFE / 10%` = 60.1 / 0.1… (a perpetuity), then discount it back like any other year.

Step 8 — Sum up and divide by shares

Add all discounted cash flows + discounted terminal value = **total equity value** ≈ **$47.1MM**. Divide by **584,000 shares** → **DCF value** ≈ **$80.6 per Nicholson share**.

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FCFE = PAT + Depreciation − Increase in Working Capital − Capex
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The DCF in numbers (first years shown)

$ MM	1972	1973	1974	1975	…	1981
Revenue (grow 6%)	58.6	62.1	65.9	69.8	…	99.0
NOP (after costs)	4.73	7.02	8.24	8.87	…	13.55
PAT (after 40% tax)	3.16	4.53	5.26	5.64	…	8.45
FCFE (cash to equity)	1.72	3.01	3.64	3.93	…	6.01
PV factor @10%	0.909	0.826	0.751	0.683	…	0.382
PV of cash flow	1.55	2.46	2.71	2.66	…	25.3*

*1981 PV includes the discounted Terminal Value. Sum of all PVs = $47.1MM equity value.

Result & what it means

DCF value ≈ $80.6 per share. Nicholson's shares trade around **$44** in the market. So the business is intrinsically worth **far more** than its market price — which tells Cooper there is room to offer a healthy premium and still get a bargain. This $80.6 is the **floor of intrinsic worth** that justifies a generous offer.

WATCH-OUTS (where students lose marks)

- **FCFE vs FCFF:** Here we compute cash to *equity* (FCFE), so we discount at **Cost of Equity**, not WACC. Matching the cash flow to the right discount rate is the #1 error.
- **Profit is not cash:** Always do Step 4 (add depreciation, subtract working-capital increase and capex). Skipping it values the wrong thing.
- **Terminal value dominates:** Notice the final discounted figure ($25.3MM) is over half the total. Small changes to the terminal assumption swing the answer a lot — state your assumption clearly.
- **Discount-rate consistency:** the model rounds Ke to 10%. Show the DDM working (D1/P + g) so the grader sees where 10% came from.

3 Method 2 — Swap Based on EPS (The Heart of the Case)

WHY this is the most important method

Cooper is paying with its **own shares**. Every new share it prints to give Nicholson holders means Cooper's existing profit is now split among more shares — which can **reduce Cooper's earnings per share (EPS)**. A falling EPS usually drags down the share price, angering Cooper's own shareholders. So Cooper asks: *"What is the **maximum** number of my shares I can give away **before my EPS starts to fall?**"* That maximum sets the **ceiling** of a safe offer.

THE BIG IDEA (plain English)

> EPS = **total profit ÷ number of shares**. When Cooper buys Nicholson, the **profit** goes up (Nicholson's profit is added) **and** the **share count** goes up (new shares issued). As long as profit rises **fast enough** to keep EPS at least where it was, the deal is not dilutive. The tipping point — where combined EPS exactly equals Cooper's old EPS — tells you the most shares Cooper can safely issue.

HOW to solve it — step by step (using 1972 numbers)

Step 1 — Cooper's stand-alone EPS

Cooper's net income = $11MM; shares = 4.22MM. Old EPS = 11 / 4.22 = **$2.607**. This is the number we must not fall below.

Step 2 — Combined profit after the deal

Add Nicholson's profit (from the DCF projection, $3.16MM) to Cooper's: 11 + 3.16 = **$14.16MM** total income available.

Step 3 — Max total shares allowed

To keep EPS at $2.607 with $14.16MM profit, total shares allowed = `14.16 / 2.607` = **5.43MM shares**.

Step 4 — Max NEW shares Cooper can issue

Cooper already has 4.22MM. So new shares it can print = 5.43 – 4.22 = **1.21MM shares**.

Step 5 — Convert to a swap ratio

Nicholson has 0.584MM shares. Swap ratio = new shares / Nicholson shares = 1.21 / 0.584 = **2.08** Cooper shares per Nicholson share.

Step 6 — Translate to value & premium

Cooper trades at $24. Value offered per Nicholson share = 2.08 × $24 = **$49.8**. Nicholson trades at $44, so this is a **13–24% premium** — attractive, yet EPS-safe.

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Max new shares = (Combined profit ÷ Cooper's old EPS) − Cooper's existing shares
Swap ratio = Max new shares ÷ Nicholson's shares
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The EPS swap across years (why the answer is a range)

Year	Combined profit ($MM)	Cooper EPS ($)	Max new shares (MM)	Swap ratio	Offer value ($)
1972	14.16	2.607	1.21	2.08	49.8
1973	16.43	2.820	1.61	2.75	66.0

1974	18.06	3.033	1.74	2.97	71.3
1975	19.44	3.270	1.73	2.95	70.9
1976	21.04	3.555	1.70	2.91	69.9

Reading it: as Nicholson's profits grow, Cooper can afford to issue more shares without dilution — so the safe swap ratio rises from ~2.1 toward ~2.9 and settles there.

WATCH-OUTS

- The logic is always: **combined profit ÷ old EPS = max total shares**. Memorise this chain; everything else follows.
- Use **Cooper's** EPS as the floor (the acquirer's), not Nicholson's. Mixing them up flips the whole answer.
- The swap ratio is a **maximum** (ceiling). Cooper can offer *less* and still be safe; it just can't offer more without hurting EPS.

4 Method 3 — Swap Based on Book Value (Cross-Check)

WHY use it

Book value = a company's **accounting net worth** (assets minus liabilities) from the balance sheet. Swapping in proportion to book value per share is a simple, objective cross-check — useful when the target's assets (like Nicholson's) are worth more than the accounts show.

HOW — step by step

Step 1 — Book value per share, both firms

Nicholson: $31MM equity / 584,000 shares = **$53.08**. Cooper: $85MM / 4.22MM = **$20.15**.

Step 2 — Swap ratio = ratio of book values

53.08 / 20.15 = **2.63** Cooper shares per Nicholson share.

Step 3 — Revalue Nicholson's real assets (the adjustment)

Nicholson's assets are understated on the books by about **$9.2MM** (real worth higher). Adjusted equity = 31 + 9.2 = $40.2MM → adjusted BV/share = **$68.84**.

Step 4 — Adjusted swap ratio

68.84 / 20.15 = **3.42**. At Cooper's $24 price that implies an offer of ~$82 — an **86% premium**.

Book Value method	Before adjustment	After revaluing assets
BV per Nicholson share	$53.08	$68.84
BV per Cooper share	$20.15	$20.15
Swap ratio	2.63	3.42
Implied offer value (× $24)	$63.2	$82.0
Premium over $44 market	44%	86%

Book value tends to give the **highest** swap ratios — treat it as an upper cross-check, not the primary answer, since accounting values can differ from economic value.

5 Method 4 — Shareholder-Group Analysis (The Negotiation Lens)

WHY it matters

A deal is not just maths — different owners must all say **yes**. A number that works on a spreadsheet fails if the family won't sell. This lens maps each group's motive to the offer feature that wins them over. It's often a full exam sub-question.

Group (shares)	What they care about	What wins them over
Family / Promoters (117,000)	Keeping some control & freedom; avoiding tax	A tax-free share exchange; a respectful long-term partner
H.K. Porter (177,000)	A clean, tax-free exit at a good price	Tax-free exchange, or cash above $50/share
Speculators (50–100k)	Highest short-run gain	A high cash bid in the open market
Non-speculators (147–197k)	Tend to follow the promoters	Whatever the family accepts

Cooper (29,000)	Control of Nicholson without VLN stock	Launch a tender offer at an attractive ratio

The design insight

The winning structure used a **convertible preferred share** paying a **$2.50 dividend**, convertible into 1.8–2.0 Cooper shares. Why clever? It gives the **family income + security** (the fixed dividend) while giving **upside** (convert to stock if Cooper rises) and is **tax-efficient** — simultaneously satisfying groups with conflicting wants.

6 Bringing the Four Methods Together

Each method gives a swap ratio in a range. You triangulate — they bracket the negotiating zone:

Method	Swap ratio	Implied value/share	Premium	Role in the answer
DCF (intrinsic worth)	—	~$80.6	high	Floor of true value
EPS non-dilutive	2.1 – 2.9	$49.8 – $71	13–38%	Ceiling for Cooper (safe zone)
Book value	2.6 – 3.4	$63 – $82	44–86%	Upper cross-check
Group analysis	—	structure-driven	—	Must win the family

The judgement

DCF says Nicholson is worth ~$80 — so a premium is justified. EPS says Cooper stays safe up to a ~2.9 ratio (~$70). Book value stretches higher. The sensible offer lands where these overlap: a swap of roughly **2 to 2.5 Cooper shares per Nicholson share**, delivered as a **convertible preferred** to win the family. That is both financially safe for Cooper and attractive to Nicholson.

What actually happened

Cooper **successfully acquired Nicholson**. It quietly bought up to ~22% of Nicholson stock in the open market (about $6.2MM), then offered the remaining holders a **convertible preferred share** paying a **$2.50 annual dividend**, convertible into roughly **1.8–2.0 Cooper shares**. The structure satisfied the family's need for income and security while protecting Cooper's EPS. Cooper's stock later reached a high of **$44** in 1973 as earnings rose — the deal created value.

7 Exam Drill — Solve Any Cooper-Type Case in 7 Moves

1. Identify the type. Stock-swap acquisition + "swap ratio" asked + shareholder groups = Cooper-type. Reach for the four lenses.

2. DCF the target. Project revenue → costs → PAT → convert to FCFE (add dep, minus ΔWC, minus capex) → discount at Cost of Equity → add terminal value → ÷ shares. Gives intrinsic floor.

3. EPS ceiling. Max total shares = combined profit ÷ acquirer's old EPS. Minus existing shares = max new. ÷ target shares = max swap ratio.

4. Book-value cross-check. BV/share target ÷ BV/share acquirer = swap ratio. Revalue hidden assets if told to.

5. Group analysis. Tabulate each shareholder group: shares, motive, what wins them. Identify who holds the swing votes (here, the family).

6. Triangulate. Overlay the ranges. Pick a ratio that is below the EPS ceiling yet above a fair premium on market price.

7. Design the instrument. If groups conflict, propose a convertible preferred (income + upside + tax efficiency) to satisfy everyone at once.

One-line memory hook

DCF sets the floor, EPS sets the ceiling, Book value is the cross-check, and the Family decides the structure. Every merger-swap case is that same sentence with different numbers.